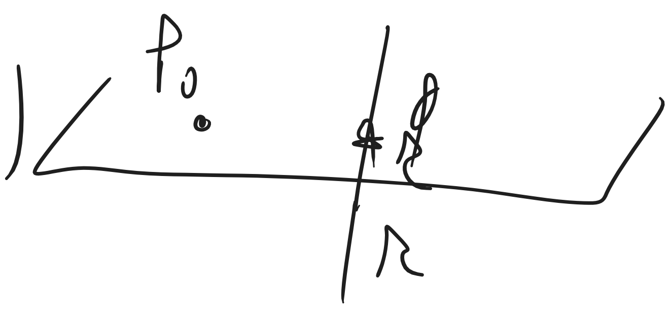
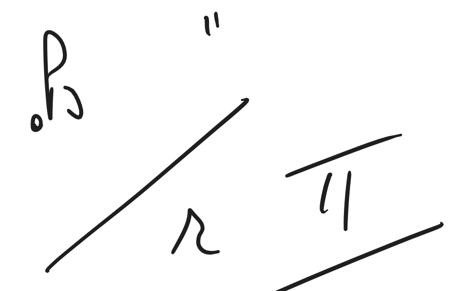
1° esercizio

dist(P_0, r)





π che contiene r e P_0

Fascio di piani generato da r

2ª esercizio: come segue

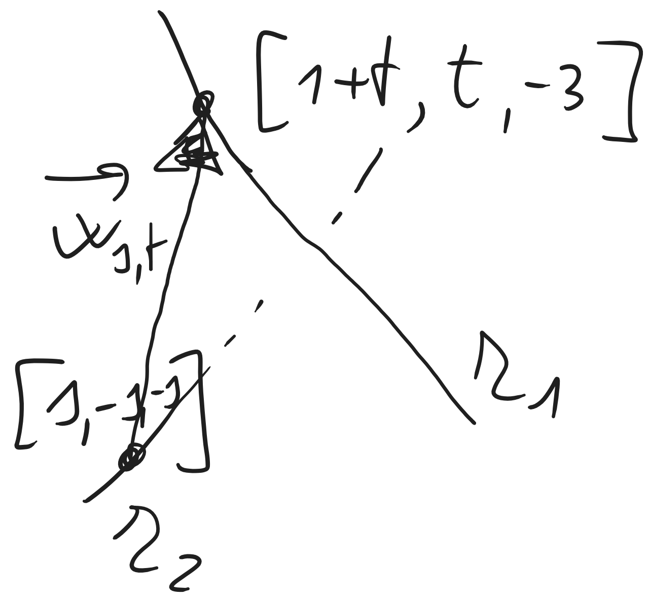
Siano $r_1: \begin{cases} x-y-1=0 \\ z+3=0 \end{cases}$ $r_2: \begin{cases} x+z=0 \\ x+y=0 \end{cases}$

Assumendo r_1, r_2 sghembe, determinare la loro comune perpendicolare r (forma parametrica e sist. lineare) precisando le coordinate dei punti di intersezione

$\underbrace{P_1 = r \cap r_1}$, $\underbrace{P_2 = r \cap r_2}$

$r_1: \begin{cases} x = 1+t \\ y = t \\ z = -3 \end{cases} \quad t \in \mathbb{R}$ $r_2: \begin{cases} x = s \\ y = -s \\ z = -s \end{cases} \quad s \in \mathbb{R}$

$r_1 \parallel \vec{w}_1 = [1,1,0]$ $r_2 \parallel \vec{w}_2 = [1,-1,-1]$



$\vec{w}_{s,t} = [1+t-s, t+s, -3+s]$

Devo imporre $\begin{cases} \vec{w}_1 \perp \vec{w}_{s,t} \\ \vec{w}_2 \perp \vec{w}_{s,t} \end{cases} \Leftrightarrow \begin{cases} \vec{w}_1 \cdot \vec{w}_{s,t} = 0 \\ \vec{w}_2 \cdot \vec{w}_{s,t} = 0 \end{cases}$

$\Leftrightarrow \begin{cases} [1,1,0] \cdot [\underbrace{1+t-s, t+s, -3+s}] = 0 \\ [1,-1,-1] \cdot [1+t-s, t+s, -3+s] = 0 \end{cases}$ $\begin{cases} 1+2t = 0 \\ 1+t-s-t-s+3-s = 0 \end{cases}$

$\begin{cases} 1+2t=0 \\ 4-3s=0 \end{cases}$ $\begin{cases} t = -1/2 \rightsquigarrow P_1 = \left[\frac{1}{2}, -\frac{1}{2}, -3\right] \\ s = 4/3 \rightsquigarrow P_2 = \left[\frac{4}{3}, -\frac{4}{3}, -\frac{4}{3}\right] \end{cases}$

$r \parallel \vec{w}_{4/3, -1/2} = \left[-\frac{5}{6}, \frac{5}{6}, \frac{-10}{6}\right] \parallel [1,-1,2]$ ↙ verifico che è $\perp \vec{w}_1, \vec{w}_2$

$r \parallel \underline{[1,-1,2]}$ e passa per $P_1 = \left[\frac{1}{2}, -\frac{1}{2}, -3\right]$

In forma parametrica, $r: \begin{cases} x = 1/2 + t \\ y = -1/2 - t \\ z = -3 + 2t \end{cases} \quad t \in \mathbb{R}$

Per trovare la forma cartesiana di r devo eliminare t:

$t = x - 1/2$

$\begin{cases} y = -1/2 - x + 1/2 \\ z = -3 + 2x - 1 \end{cases}$ $r: \begin{cases} x+y=0 \\ 2x-z-4=0 \end{cases}$ O.K.

↗ verificare che P_1 e P_2 soddisfano